

SYNOPSIS: This Sources Sought Announcement is to assist the US Army Combat Capabilities Development Command - Soldier Center (DEVCOM-SC) to identify potential sources to provide information for the next-generation Insulated Food Containers and Insulated Beverage Containers with improved thermal performance, collapsible design for reduced logistics footprint, and optional active thermal management capability. The IFC-i is intended to replace CID A-A-52193G and the IBC-i is intended to replace CID A-A-52190C for military expeditionary feeding operations supporting UGR-H&S ration transport from central preparation to forward feeding sites.

The Government requests responders submit a Broad Other Transaction Authority Announcement (BOTAA) preproposal as outlined in DEVCOM-SC BOTAA 22-01 located at: <https://sam.gov/opp/126eaf3a78d847f69b7c94b606877f92/view>. Please refer to Topic Area of Interest "A. SOLDIER SUSTAINMENT DIRECTORATE (SSD)" starting on page 19. Submit the BOTAA preproposal to: james.d.nelson196.civ@army.mil and jean.m.trumpis.civ@army.mil no later than 08 May 2026 by 5PM EDT.

Objective: DEVCOM-SC is seeking information on technologies or capabilities to provide highly mobile, scalable, modular insulated food and beverage storage and transport solutions supporting distributed feeding operations in austere environments. The IFC-i/IBC-i shall provide collapsible, lightweight, energy efficient containers with passive and optional active hot/cold capabilities, reducing logistical footprint while maintaining food safety ($\geq 140^{\circ}\text{F}$ hot hold, $\leq 41^{\circ}\text{F}$ cold hold per HACCP/ServSafe).

Program Alignment: This effort supports replacement of current IFCs/IBCs and integration with fielded platforms (Tray Ration Heating Systems, Expeditionary Field Kitchen, Assault Kitchen, Containerized Kitchen, Mobile Kitchen Trailer) and future systems (Expeditionary Field Feeding Equipment System, Scalable Combat Feeding Platform).

Background:

DEVCOM-SC Combat Feeding Division has identified the need for improved insulated food and beverage containers to support expeditionary feeding operations. Current Insulated Food Containers (IFCs) and Insulated Beverage Containers (IBCs) are heavy, occupy the same volume empty as full, and exhibit performance limitations under field serving conditions. This initiative supports the Army's modernization goals for a smaller, lighter, more mobile force operating in contested logistics environments.

Requirements:

The purpose of this announcement is to request preproposals on technologies or capabilities to provide an Improved Insulated Food Container (IFC-i), and an Improved Insulated Beverage Container (IBC-i). Respondents should demonstrate capability to address one or more of the following requirement areas:

Specific Specifications:

1. Passive Thermal Performance Requirements

The IFC-i shall meet the following thermal performance thresholds without active heating/cooling or PCM:

Test Condition	Threshold	Objective	Final Temp
Hot Hold: 180°F ±2°F initial, -20°F ±5°F ambient, 4 hours	≤40°F loss	≤35°F loss	≥140°F
Cold Hold: 33°F +2/-0°F initial, +120°F ±5°F ambient, 4 hours	≤8°F rise	≤6°F rise	≤41°F
Extended Hold (with PCM)	≥5 hours	≥6 hours	Per above
<i>Extreme Cold (Stretch): 180°F initial, -60°F ambient</i>	<i>≥4 hrs to 140°F</i>	<i>≥6 hrs to 140°F</i>	<i>≥140°F</i>

Extreme Cold Stretch Goal: Per program objectives, respondents with solutions capable of maintaining ≥140°F for 5-6 hours at -60°F ambient are encouraged to provide supporting test data.

2. Insulation Requirements

Parameter	Threshold	Objective	Baseline
Effective R-value (per inch)	≥7.0	≥12.0	~4.0
Total wall thickness (insulation + shell)	≤1.5"	≤1.25"	~2.0"

Acceptable insulation technologies include but are not limited to: Flexible aerogel composites, multi-layer reflective insulation (MLI), closed-cell polyurethane foam, vacuum insulated panels (VIPs), microencapsulated PCM-embedded materials, functional polymers, or hybrid systems. Technologies enabling high thermal resistivity within flexible, durable structures are of particular interest for collapsible designs.

Other Goals: Respondents with advanced insulation technologies achieving R-10.5 to R-12 per inch (per program goals) while maintaining collapsibility and durability are encouraged to respond, with supporting test data.

3. Serving-Line Performance Test

Because real-world performance degrades when lid is opened for serving, the IFC-i shall pass a serving simulation test:

1. Load container per standard hot test protocol (180°F water)
2. Conduct 25 open/close cycles over 30 minutes (simulating serving line)
3. Each cycle: open lid, wait 15 seconds, close lid, wait 45 seconds
4. After serving simulation, continue closed hold test
5. **Pass criterion: Contents remain ≥140°F for ≥90 minutes after serving simulation completes**

4. Collapsibility / Reduced Logistics Footprint

The IFC-i and IBC-i shall reduce stowed volume when empty to minimize logistics burden. Current rigid containers occupy the same volume empty as full, creating transportation inefficiencies.

Insulated Food Container - Improved		
Parameter	Threshold	Objective
Volume reduction (collapsed vs. deployed)	≥25%	≥30%
Operational dimensions (deployed)	≤17 ⁷ / ₈ " W × 25 ¹ / ₈ " L × 10 ¹ / ₄ " H	
Collapse/deploy cycle endurance	≥200 cycles, no degradation	
Insulated Beverage Container - Improved		
Volume reduction (collapsed vs. deployed)	≥25%	≥30%
Operational dimensions (deployed)	≤9" W × 16 ³ / ₄ " L × 23 ⁷ / ₈ " H	
Collapse/deploy cycle endurance	≥200 cycles, no degradation	

Acceptable Design Approaches:

- Collapsible rigid frame with hinged panels or telescoping structure
- Thin-walled fabric super-insulated shell coupled to collapsible frame
- Removable/modular insulation systems
- Nesting designs for multiple unit storage
- Flexible durable textile constructions (CFD has in-house textile fabrication capability for soft prototype development)

Design Constraint: Collapsibility shall be achieved through mechanical design rather than compressible insulation. System shall return to full rigidity and thermal sealing capability upon deployment. Collapsible elements shall lock positively in deployed position with no play or flex.

5. Phase Change Material (PCM) Integration

The IFC-i shall accept removable PCM modules to extend thermal hold duration:

PCM Type	Melt Point	Function
Hot Hold PCM	145–160°F	Releases heat at danger zone boundary
Cold Hold PCM	-20–35°F	Absorbs heat near freezing point

PCM module requirements: Field-replaceable, washable, durable to -40°F storage, no leakage when damaged.

6. Physical and Weight Requirements

Condition	Threshold	Objective
Empty weight (container + inserts + lids)	≤25 lbs	≤20 lbs

Condition	Threshold	Objective
Loaded weight (food + PCM if installed)	≤75 lbs	
Active PCM module (optional, if installed)	≤10 lbs	≤8 lbs
Feeding capacity (per module)	25 personnel	50 (2 modules)
Stackability	4 containers high	5 containers high

Two-person carry requirement: Container shall include integrated handles permitting gloved two-person lift and carry at maximum loaded weight. Handles operable with arctic gloves.

7. Insert Pan Configuration

The IFC-i interior shall accommodate the following insert configurations, matching UGR-H&S tray pack requirements:

- a. **Primary:** Three (3) ½-size, 6" deep insert pans with sealed lids and gaskets
- b. **Alternate:** One (1) full-size, 6" deep pan with sealed lid and gasket
- c. **Tray Pack:** Three (3) tray pack cans (Size 1 baseline) or four (4) tray pack cans (Size 2 extended)

8. Cleanability

The following shall be incorporated into the final IFC-i and IBC-i design:

- a. **Materials:** All food-contact surfaces are fabricated from National Sanitation Foundation (NSF) certified non-porous, non-toxic, corrosion-resistant, and non-absorbent materials that will not impart odor, color, or taste to food.
- b. **Surface Finish:** Food-contact surfaces are smooth, durable, and free of pits, cracks, crevices, and sharp internal angles where food particles and microorganisms can be harbored. Each internal corners shall be rounded to a sufficient radius to permit easy cleaning.
- c. **Accessibility and Disassembly:** All food-contact surfaces shall be readily accessible for manual cleaning and inspection, and while disassembly is not encouraged, if disassembly is necessary for cleaning then this shall be completed without the need for specialized tools.
- d. **Drainage:** The Equipment and any of its removable parts shall be self-draining or can be easily tilted or inverted for complete draining, preventing the pooling of liquids that can lead to microbial growth.
- e. **Industry Standards:** The Equipment shall be designed and constructed to meet, at a minimum, the relevant NSF/ANSI standards for food equipment cleanability applicable at the time of manufacture.

9. Technical Data Package

The contractor shall develop, maintain, and deliver a Product Level Technical Data Package (TDP) in accordance with MIL-STD-31000C.

The TDP shall be sufficient to enable competitive procurement and life-cycle sustainment. At a minimum, the TDP shall include:

- a. 3D native CAD models and associated 2D drawings in PDF format.
- b. A complete Bill of Materials (BOM) identifying all parts, materials, and their sources.
- c. All material and process specifications used in manufacturing.
- d. Special Packaging Instructions.

10. Active Thermal Management (Optional Enhanced Capability)

Note: Active heating/cooling is an optional enhanced capability. The baseline IFC-i shall meet all passive requirements without active systems. Trade-off analysis will determine cost-benefit for fielding.

Parameter	Threshold	Objective	Stretch Goal
Heating capacity	50W	350W	1035W
Cooling capacity (TEC-based)	50W	150W	1035W
Battery runtime (integrated)	≥2 hours	≥4 hours	—
Power interfaces	12V DC	12V DC + 120V AC	+ Solar/Wind

Design Requirement: Active module shall be removable as single unit for repair/replacement. Dual-function (heating/cooling within same footprint) capability is desired.

11. Durability and Environmental Requirements

Test	Requirement
Drop test	36" drop onto concrete, loaded, all orientations—no structural failure
Vibration test	MIL-STD-810H Method 514.8, Category 4 (wheeled vehicle)
Latch endurance	≥500 cycles, no deformation or loss of clamping force
Collapse/deploy cycle	≥200 cycles, no degradation of thermal performance or seal integrity
Operating temperature	-20°F to +120°F operating; -60°F to +140°F storage
Taber Abrasion	Exterior surfaces shall show no exposure of substrate material when tested per ASTM D4060 (CS-17 wheels, 1000g load). Threshold: ≥2,000 cycles. Objective: ≥4,000 cycles.

12. Regulatory and Safety Compliance

- NSF/ANSI Standard 2 compliant for food equipment (listing required prior to first shipment)
- FDA 21 CFR compliant for food contact materials
- Hot food maintained $\geq 140^{\circ}\text{F}$ per HACCP/ServSafe guidelines
- Cold food maintained $\leq 41^{\circ}\text{F}$ per HACCP/ServSafe guidelines
- No chemical leaching, off-flavors, or contamination from materials
- Color options: Brown (Desert Sand) or Green (Olive) per FED-STD-595

13. Technologies of Interest

Based on Combat Feeding Division research priorities, the Government is particularly interested in the following technology areas:

- Advanced Insulation Materials:** Functional polymers and microencapsulation technologies enabling lightweight, flexible, high-performance insulation; embedded polymers and coatings offering high thermal resistivity within flexible, durable webs
- Collapsible Frame/Structure Technologies:** Collapsible frames or structures coupled with thin-walled fabric super-insulated shells; hinged panels, telescoping frames, removable insulation modules
- Phase Change Materials:** Hot/cold plates or PCM inserts for passive hold extension; PCM-embedded insulation materials tuned for expeditionary conditions; flexible insulated blankets with integrated PCM
- Active Thermal Systems:** Silicone heating pads, thermoelectric cooling modules, compact battery-powered heating/cooling; dual-function heating/cooling within single footprint
- Surface Treatments:** Solar absorption/reflectance paints and coatings for passive thermal management
- Alternative Power Sources:** Solar, wind, and fuel cell integration for active system power; vehicle power interface optimization
- Water Heating Capability:** Methods to heat water for beverages/coffee using IBC-i; hot water for reconstituting freeze-dried or condensed foods at point of consumption

Interested parties are invited to submit a response to this Sources Sought Announcement. THIS IS A SOURCES SOUGHT ANNOUNCEMENT ONLY

This Sources Sought Announcement is issued solely for information and planning purposes and to identify interested sources. THIS IS NOT A SOLICITATION. No contract will be awarded from this announcement. This Sources Sought does not constitute a Request for Proposal (RFP) or a promise to issue an RFP in the future. It is subject to change and is not binding on the Government. Further, unsolicited proposals will not be accepted. Funding is not available at this time. The United States Army has not made a commitment to procure any of the items/services discussed, and release of this Sources Sought Announcement should not be construed as such a commitment or as authorization to incur cost for which reimbursement would be required or sought. Response to this Sources Sought Announcement is voluntary and no reimbursement will be made for any costs associated with providing information in response to this and any follow-on information requests. All submissions become Government property and will not be returned.

Not responding to this Sources Sought Announcement does not preclude participation in any future RFP if any is issued. If a solicitation is released, it will be synopsized on the SAM.gov website. It is the responsibility of the potential responders to monitor this site for additional information pertaining to this subject.

RESPONSES:

Interested parties may identify their interest and capability by sending BOTAA preproposals regarding this requirement to **DEVCOM-SC via e-mail** ONLY to james.d.nelson196.civ@army.mil and jean.m.trumpis.civ@army.mil no later than 08 May 2026 by 5PM EDT. The U.S. Government will not pay for any information or administrative cost incurred in response to this Notice. All costs associated with responding to this Notice will be solely at the expense of the interested party.

Please provide business size (indicate your socioeconomic status), applicable NAICS code, and CAGE code.

QUESTIONS:

Any questions for clarification may be emailed to james.d.nelson196.civ@army.mil and jean.m.trumpis.civ@army.mil no later than 05 May 2026 by 5PM EDT. Verbal questions will NOT be accepted. Questions shall NOT contain proprietary or classified information. An unattributed list of questions and answers will be published at the same web location of this Sources Sought Announcement.